Latex 公式

(开始时间：2019/12/08)

(by 亲持红叶)

1 CATLOG

1 CATLOG
2 一、基础
2.1 、一元一次方程
2.2 、上下标
2.3 、求和
2.4 、分数
2.5 、矩阵
2.6 、希腊字母
2.7 、省略号
2.8 、大括号
2.9 、大大括号
2.10 、多行公式
2.11 补充
3 二、数学公式
3.1 、傅里叶级数
3.2 、高斯公式
3.3 、定积分
3.4 、和的展开式
3.5 、三角恒等式
3.6 、欧拉公式
3.7 、格林公式
3.8 、伯努利方程
3.9 、全微分方程
3.10 、非齐次方程
3.11 、柯西中值定理
3.12 、拉格朗日中值定理
3.13 、曲面面积
3.14 、重积分
3.15 、arcsinx 求导
3.16 、泰勒级数
3.17 、三角函数积分
3.18 、二次曲面
3.19 、二阶微分
3.20 、方向导数
4 三、公式
4.1 、练习一
4.2 、练习二
4.3 、练习三
4.4 、练习四
4.5 其他

2 一、基础

2.1 、一元一次方程

```
$$
y = a+bx
$$
```

$$y = a + bx \tag{1}$$

2.2 、上下标

```
$$
y = a + b_{ij}^2
$$
```

$$y = a + b_{ij}^2 \tag{2}$$

2.3 、求和

```
$$
\sum_{i=0}^n(x_i^2+y_j^3)
$$
```

$$\sum_{i=0}^n(x_i^2+y_j^3) \tag{3}$$

2.4 、分数

```
$$
\frac{1}{3}-\frac{x_i^2}{y_j^3}
$$

或者

$$
\cfrac{1}{3}-\cfrac{x_i^2}{y_j^3}
$$
```

$$\begin{aligned}\frac{1}{3}-\frac{x_i^2}{y_j^3}\\ \frac{1}{3}-\frac{x_i^2}{y_j^3}\end{aligned} \tag{4}$$

2.5 、矩阵

```
$$
\left[
\begin{array} {cccc}
X_1&Y_1^2\\
X_2 & Y_2^2\\
\ldots \\
X_n & Y_n^2
\end{array}
\right]
$$
```

$$\left[\begin{array} {cccc}X_1&Y_1^2\\X_2 & Y_2^2\\\ldots \\X_n & Y_n^2\end{array}\right] \tag{5}$$

2.6 、希腊字母

- 首字母是否大写是区分希腊字母大小写的关键

小写	大写	Latex小写命令	Latex大写命令
α	A	\alpha	\Alpha
β	B	\beta	\Beta
γ	Γ	\gamma	\Gamma
δ	Δ	\delta	\Delta
ϵ	E	\epsilon	\Epsilon
ζ	Z	\zeta	\Zeta
ν	N	\nu	\Nu
ξ	Ξ	\xi	\Xi
o	O	\omicron	\Omicron
π	Π	\pi	\Pi
ρ	P	\rho	\Rho
σ	Σ	\sigma	\Sigma
η	H	\eta	\Eta
θ	Θ	\theta	\Theta
ι	I	\iota	\Iota
κ	K	\kappa	\Kappa
λ	Λ	\lambda	\Lambda
μ	M	\mu	\Mu
τ	T	\tau	\Tau
υ	Υ	\upsilon	\Upsilon
ϕ	ϕ	\Phi	\Phi
χ	X	\chi	\Chi
ψ	Ψ	\psi	\Psi
ω	Ω	\omega	\Omega

2.7 、省略号

```
-- \ldots   与底线对齐的省略号，\cdots 与中线对齐省略号
-- 换行 "\\"

$$
f(x_1,x_2,\ldots,x_i) = x_1+x_2+\cdots+x_i
$$

```

$$f(x_1,x_2,\ldots,x_i) = x_1+x_2+\cdots+x_i \tag{6}$$

$$\begin{aligned} & f(x_1,x_2,\ldots,x_i) \\ & = x_1+x_2+\cdots+x_i \end{aligned} \tag{7}$$

2.8 、大括号

```
" \{  \}""  (注意这里直接{}打是不行的,显示不出来的)

$$
\{a+b\}
$$
```

$$\{a+b\} \tag{8}$$

2.9 、大大括号

```
* "\left(  \right)"
* "\left[  \right]"

$$
\left(a+b^2+(c*r)\right)
$$

或者

$$
\left[a+b^2+(x*r)\right]
$$
```

$$\left(a+b^2+(c*r)\right)$$
$$\left[a+b^2+(x*r)\right] \tag{9}$$

2.10 、多行公式

```
$$
\begin{eqnarray*}
\cos 2 \theta & = & \cos^2 \theta - \sin^2 \theta\\
&=& 2\cos^2 \theta - 1
\end{eqnarray*}
$$

或者

$$
\begin{eqnarray*}
\cos 2 \theta & = & \cos^2 \theta - \sin^2 \theta\\
&=& 2\cos^2 \theta - 1
\end{eqnarray*}
$$

```

$$\begin{eqnarray*}
\cos 2 \theta & = & \cos^2 \theta - \sin^2 \theta\\
&=& 2\cos^2 \theta - 1
\end{eqnarray*}$$

或者

$$\begin{eqnarray*}
\cos 2 \theta & = & \cos^2 \theta - \sin^2 \theta\\
&=& 2\cos^2 \theta - 1
\end{eqnarray*}$$

2.11 补充

- 对后续使用的符号的补充

意思	Latex	显示	备注
大于等于	\geq	$\geq$	greater or equal to
小于等于	\leg	$\leq$	less than or equal to
约等于	\approx	$\approx$	approximately equal to
属于、不属于	\in,\notin	$\in, \notin$	
因为、所以	\because,\therrefore	$\because, \therefore$	
正比于	\propto	$\propto$	Proportional to
恒等、求和、连乘	\equiv,\sum,\prod	$\equiv, \sum, \prod$	
全称、存在	\forall,\exists	$\forall, \exists$	
角度	\ang	$\angle$	Angel
趋于	\rightarrow	$\rightarrow$	
无穷大	\infty	∞	

3 二、数学公式

- 下面的公式是从wps中套用过来的

3.1 、傅里叶级数

```
$$
f(x) = \frac{a_0}{2} + \sum_{n=1}^\infty(a_n \cos {nx} + b_n \sin {nx})
$$

或者

$$
\{ f(x) = {{{a_0}} \over 2} + \sum\limits_{n = 1}^\infty  {({a_n}\cos {nx} + {b_n}\sin {nx})}  \}
$$
```

$$
\begin{aligned}
f(x) = \frac{a_0}{2} + \sum_{n=1}^\infty(a_n \cos {nx} + b_n \sin {nx}) \\
\{f(x) = \frac{a_0}{2} + \sum_{n=1}^\infty(a_n \cos {nx} + b_n \sin {nx})\}
\end{aligned} \tag{12}
$$

3.2 、高斯公式

```
$$
\iiint _ { \Omega } \left( \frac { \partial {P} } { \partial {x} } + \frac { \partial {Q} } { \partial {y} } + \frac { \partial {R} }{ \partial {z} } \right) \mathrm { d } V = \oint _ { \partial \Omega } ( P \cos \alpha + Q \cos \beta + R \cos \gamma ) \mathrm{ d} S
$$
```

$$
\iiint _ { \Omega } \left(\frac { \partial {P} } { \partial {x} } + \frac { \partial {Q} } { \partial {y} } + \frac { \partial {R} }{ \partial {z} } \right) \mathrm { d } V = \oint _ { \partial \Omega } (P \cos \alpha + Q \cos \beta + R \cos \gamma) \mathrm{ d} S \tag{13}
$$

3.3 、定积分

```
$$
\lim\limits_ { n \rightarrow + \infty } \sum _ { i = 1 } ^ { n } f \left[ a + \frac { i } { n } ( b - a ) \right] \frac { b - a } { n } = \int _ { a } ^ { b } f ( x )\mathrm {d}x
$$
```

$$
\lim\limits_ { n \rightarrow + \infty } \sum _ { i = 1 } ^ { n } f \left[a + \frac { i } { n } (b - a) \right] \frac { b - a } { n } = \int _ { a } ^ { b } f (x)\mathrm {d}x \tag{14}
$$

3.4 、和的展开式

```
$$
( 1 + x ) ^ { n } = 1 + \frac { n x } { 1 ! } + \frac { n ( n - 1 ) x ^ { 2 } } { 2 ! } + \cdots
$$
```

$$
(1 + x) ^ { n } = 1 + \frac { n x } { 1 ! } + \frac { n (n - 1) x ^ { 2 } } { 2 ! } + \cdots \tag{15}
$$

3.5 、三角恒等式

```
$$
\sin \alpha \pm \sin \beta = 2 \sin \frac { 1 } { 2 } ( \alpha \pm \beta ) \cos \frac { 1 } { 2 } ( \alpha \mp \beta )
$$



$$
\cos \alpha + \cos \beta = 2 \cos \frac { 1 } { 2 } ( \alpha + \beta ) \cos \frac { 1 } { 2 } ( \alpha - \beta )
$$
```

$$
\begin{aligned}
\sin \alpha \pm \sin \beta = 2 \sin \frac { 1 } { 2 } (\alpha \pm \beta) \cos \frac { 1 } { 2 } (\alpha \mp \beta) \\
\cos \alpha + \cos \beta = 2 \cos \frac { 1 } { 2 } (\alpha + \beta) \cos \frac { 1 } { 2 } (\alpha - \beta)
\end{aligned} \tag{16}
$$

3.6 、欧拉公式

```
$$
{e^{ix}} = \cos {x} + i\sin {x}
$$
```

$$
e^{ix} = \cos x + i \sin x \tag{17}
$$

3.7 、格林公式

```
$$
\int\!\!\!\int\limits_D {({{\partial Q} \over {\partial x}} - {{\partial P} \over {\partial y}})dxdy = \oint\limits_L {Pdx + Qdy} }
$$
```

$$
\iint\limits_D ({\frac{\partial Q}{\partial x}} - {\frac{\partial P}{\partial y}})dxdy = \oint\limits_L {Pdx + Qdy} \tag{18}
$$

3.8 、伯努利方程

```
$$
{{dy} \over {dx}} + P(x)y = Q(x){y^n}(n \ne 0,1)
$$
```

$$
{\frac{dy}{dx}} + P(x)y = Q(x){y^n}(n \ne 0,1) \tag{19}
$$

3.9 、全微分方程

```
$$
du(x,y) = P(x,y)dx + Q(x,y)dy = 0
$$
```

$$
du(x,y) = P(x,y)dx + Q(x,y)dy = 0 \tag{20}
$$

3.10 、非齐次方程

```
$$
y = (\int {Q(x){e^{\int {P(x)dx} }}dx + C){e^{ - \int {P(x)dx} }}}
$$
```

$$y = (\int Q(x)e^{\int P(x)dx}dx + C)e^{-\int P(x)dx} \tag{21}$$

3.11 、柯西中值定理

```
$$
\frac{{f(b) - f(a)}}{{F(b) - F(a)}} = \frac{{f'(\xi )}}{{F'(\xi )}}
$$
```

$$\frac{f(b) - f(a)}{F(b) - F(a)} = \frac{f'(\xi)}{F'(\xi)} \tag{22}$$

3.12 、拉格朗日中值定理

```
$$
f(b) - f(a) = f'(\xi )(b - a)
$$
```

$$f(b) - f(a) = f'(\xi)(b - a) \tag{23}$$

3.13 、曲面面积

```
$$
A = \int\!\!\!\int\limits_D {\sqrt {1 + {{\left( {{{\partial z} \over {\partial x}}} \right)}^2} + {{\left( {{{\partial z} \over {\partial y}}} \right)}^2}} dxdy}
$$
```

$$A = \iint\limits_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dxdy \tag{24}$$

3.14 、重积分

```
$$
\int\!\!\!\int\limits_D {f(x,y)dxdy}  = \int\!\!\!\int\limits_{D'} {f(r\cos \theta ,r\sin \theta )rdrd\theta }
$$
```

$$\iint\limits_D f(x,y)dxdy = \iint\limits_{D'} f(r\cos\theta, r\sin\theta)rdrd\theta \tag{25}$$

3.15 、arcsinx 求导

```
$$
(\arcsin x)' = \frac{1}{{\sqrt {1 - {x^2}} }}
$$
```

$$(\arcsin x)' = \frac{1}{\sqrt{1 - x^2}} \tag{26}$$

3.16 、泰勒级数

```
$$ p;ko[p';p[oooo]]
e^x = 1 + \frac{x}{1!} +  \frac{x}{2!}+ \frac{x}{3!} + \cdots + O(x^n),     -\infty < x < \infty
$$
```

$$e^x = 1 + \frac{x}{1!} + \frac{x}{2!} + \frac{x}{3!} + \cdots + O(x^n), -\infty < x < \infty \tag{27}$$

3.17 、三角函数积分

```
$$
\int tgx \mathrm {d} x = -\ln {|\cos x|} + c
$$
```

$$\int tgx \mathrm{d} x = -\ln {|\cos x|} + c \tag{28}$$

3.18 、二次曲面

```
$$
\frac{{{x^2}}}{{{a^2}}} + \frac{{{y^2}}}{{{b^2}}} - \frac{{{z^2}}}{{{c^2}}} = 1
$$
```

$$\frac{{{x^2}}}{{{a^2}}} + \frac{{{y^2}}}{{{b^2}}} - \frac{{{z^2}}}{{{c^2}}} = 1 \tag{29}$$

3.19 、二阶微分

```
$$
\frac{d^2y}{dx^2}+P(x)\frac{dy}{dx}+Q(x)y=f(x)
$$
```

$$\frac{d^2y}{dx^2}+P(x)\frac{dy}{dx}+Q(x)y=f(x) \tag{30}$$

3.20 、方向导数

```
$$
\frac{\partial f}{\partial l}=\frac{\partial f}{\partial x}\cos{\phi}+\frac{\partial f}{\partial y}\sin{\phi}
$$
```

$$\frac{\partial f}{\partial l}=\frac{\partial f}{\partial x}\cos{\phi}+\frac{\partial f}{\partial y}\sin{\phi} \tag{31}$$

4 三、公式

- 下面的公式来自清风数学建模中有一张交latex给出的4个

4.1 、练习一

```
$$
\min {f_1} = \frac{\sqrt{\frac{1}{n-1}\sum(W_j-\overline{W}^2)}}{\overline{W}}

\\

\min{f_2}= \max\limits_{1<j<n}{T_j}

\\

\operatorname{ s.t. }

\left\{
\begin{array}{}

x_{ij} = 1, i = j \\

\sum\limits_{j=1}^n x_{ij}=1, i \neq j \\

\frac{S_{ij}}{V} \leq 3\\

x_{ij}  \in \{0,1\} \\

\overline{W} = \frac{1}{n}\sum\limits_{j=1}^n W_j\\

(i = 1,2,3,\cdots,m;j=1,2,\cdots,n)

\end{array}
\right.
$$
```

$$
\begin{aligned}
&\min f_1 = \frac{\sqrt{\frac{1}{n-1}\sum(W_j-\overline{W}^2)}}{\overline{W}} \\
&\min f_2 = \max\limits_{1<j<n} T_j \\
&\operatorname{ s.t. }\left\{\begin{array}{c}
x_{ij} = 1, i = j \\
\sum\limits_{j=1}^n x_{ij}=1, i \neq j \\
\frac{S_{ij}}{V} \leq 3 \\
x_{ij} \in \{0,1\} \\
\overline{W} = \frac{1}{n}\sum\limits_{j=1}^n W_j \\
(i = 1,2,3,\cdots,m;j=1,2,\cdots,n)
\end{array}\right.
\end{aligned} \tag{32}
$$

4.2 、练习二

```
$$
\begin{eqnarray*}
\int_{k-1}^k \frac{dx_0^{(1)}(t)}{dt}\mathrm{dt} & \approx & -\hat{a}\int_{k-1}^kx_0^{(1)}dt+\int_{k-1}^k\hat b dt

\\

& = & \int_{k-1}^k \left[-\hat a x_0 ^ {(1)} (t) + \hat b \right]dt

\end{eqnarray*}
$$
```

$$
\begin{eqnarray*}
\int_{k-1}^k \frac{dx_0^{(1)}(t)}{dt}\mathrm{dt} & \approx & -\hat{a}\int_{k-1}^kx_0^{(1)}dt+\int_{k-1}^k\hat b dt

\\

& = & \int_{k-1}^k \left[-\hat a x_0 ^ {(1)} (t) + \hat b \right]dt

\end{eqnarray*}
$$

4.3 、练习三

```
$$
\left\{
\begin{eqnarray*}
z_1 & = & l_{11}x_1 + l_{12}x_2 + \cdots  +  l_{1p}x_p
\\
z_2 & = & l_{21}x_1 + l_{22}x_2 + \cdots  +  l_{2p}x_p
\\
& \vdots &
\\
z_m & = & l_{m1}x_1 + l_{m2}x_2 + \cdots  +  l_{mp}x_p
\end{eqnarray*}
\right.
$$
```

$$
\left\{
\begin{eqnarray*}
z_1 & = & l_{11}x_1 + l_{12}x_2 + \cdots + l_{1p}x_p
\\
z_2 & = & l_{21}x_1 + l_{22}x_2 + \cdots + l_{2p}x_p
\\
& \vdots &
\\
z_m & = & l_{m1}x_1 + l_{m2}x_2 + \cdots + l_{mp}x_p
\end{eqnarray*}
\right.
$$

4.4 、练习四

```
$$
x =
\left[
\begin{array}{随便写点啥，反正不能不写}
 x_{11} & x_{12} & \cdots & x_{1p} \\
 x_{21} & x_{22} & \cdots & x_{2p} \\
 \vdots & \vdots & \ddots & \vdots \\
x_{n1} & x_{n2} & \cdots & x_{np} \\
\end{array}
\right]
=(x_1,x_2,\cdots,x_p)
$$
```

$$
x =
\left[
\begin{array}{随便写点啥，反正不能不写}
 x_{11} & x_{12} & \cdots & x_{1p} \\
 x_{21} & x_{22} & \cdots & x_{2p} \\
 \vdots & \vdots & \ddots & \vdots \\
x_{n1} & x_{n2} & \cdots & x_{np} \\
\end{array}
\right]
=(x_1,x_2,\cdots,x_p) \tag{1}
$$

$$y = f(x) = W^T \cdot x + b = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b \tag{2}$$

4.5 其他

$$\chi_A = \begin{cases} 1, & \text{A occurs ,} \\ 0, & \text{A doesn't occur.} \end{cases} \tag{3}$$

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \tag{4}$$

$$\hat{\omega}_i = \frac{\sqrt[n]{\prod\limits_{j=1}^{n} a_{ij}}}{\sum\limits_{j=1}^{n} \sqrt[n]{\prod\limits_{j=1}^{n} a_{ij}}}, i \in [1, m] \qquad \text{(m为指标数量)}$$

$$CR = \frac{CI}{RI} = \frac{\lambda_{max} - n}{RI(n-1)} \tag{5}$$

$$B_j = -\sum_{i=1}^{m} p_{ij} \frac{\ln p_{ij}}{\ln m} \tag{6}$$

$$w_j = \frac{1 - B_j}{\sum\limits_{j=1}^{n} (1 - B_j)} \tag{7}$$

$$\sum_{j=1}^{n} w_j = 1, j \in [1, n] \tag{8}$$

$$\overline{\omega} = \left\{ \frac{W_1 w_1}{\sum_{j=1}^{n} W_j w_j}, \frac{W_2 w_2}{\sum_{j=1}^{n} W_j w_j}, \cdots, \frac{W_n w_n}{\sum_{j=1}^{n} W_j w_j} \right\} = (\omega_1, \omega_2, \cdots, \omega_n) \tag{9}$$

$$\text{s.t.} \sum_{j=1}^{n} \omega_j = 1; \omega_j > 0 \tag{10}$$

$$w_j (j = 1, 2, \cdots, n) \tag{11}$$

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix} \tag{12}$$

那么，对其标准化的矩阵记为Z， Z 中的每一个元素:

$$z_{ij} = x_{ij} / \sqrt{\sum_{i=1}^{n} x_{ij}^2} \tag{13}$$

$$\begin{aligned} Z^- &= \left(Z_1^-, Z_2^-, \cdots, Z_m^-\right) \\ &= \left(\min\{z_{11}, z_{21}, \cdots, z_{n1}\}, \min\{z_{12}, z_{22}, \cdots, z_{n2}\}, \cdots, \min\{z_{1m}, z_{2m}, \cdots, z_{nm}\}\right) \end{aligned} \tag{14}$$

$$D_i^+ = \sqrt{\sum_{j=1}^{m}(Z_j^+ - z_{ij})^2} \tag{15}$$

$$S_i = \frac{D_i^-}{D_i^+ + D_i^-} S_i \in [0,1] \tag{16}$$

$$\tilde{S}_i = \frac{S_i}{\sum\limits_{i=1}^{n} \tilde{S}_i} \tag{17}$$

$$\sum_{i=1}^{n} \tilde{S}_i = 1 \tag{18}$$

$$\min f_1 = \frac{\sqrt{\frac{1}{n-1}\sum\limits_{j=1}^{n}(W_j - \overline{W})^2}}{\overline{W}} \tag{19}$$

$$\min f_2 = \max_{1<j<n} T_j \tag{20}$$

$$\int_{k-1}^{k} \frac{dx_0^{(1)}(t)}{dt} dt \approx -\hat{a} \int_{k-1}^{k} x_0^{(1)}$$

$$\begin{gathered}
1)\text{夜间经济指数} = [\text{夜间经济企业活跃度} \times \tfrac{1}{3} + \text{夜间经济企业达产率\$} \times \tfrac{1}{3} + \text{\$夜间经济比重提升率\$} \times \tfrac{1}{3}] \times 100\text{\$} \\
(2)\text{夜间经济企业活跃度\$} = \text{\$夜间经济企业活跃度\$} \div \text{\$夜间经济企业总数} \\
(3)\text{夜间经济企业达产率\$} = (\text{\$当日夜间经济电量\$} \div \text{\$基准夜间电量\$}) \times 100\%\text{\$} \\
(4)\text{夜间经济比重提升率\$} = \text{\$夜间用电量占比\$} \div \text{\$去年同月平均夜间用电量占比} \\
(5)\text{夜间用电量占比\$} = \text{\$夜间用电量\$} \div \text{\$当日总用电量}
\end{gathered} \tag{21}$$

$$\begin{gathered}
f_1 = \frac{\sqrt{\frac{1}{n-1}\sum\limits_{j=1}^{n}(W_j - \overline{W})^2}}{\overline{W}} \\
\min f_2 = \max_{1<j<n} T_j
\end{gathered} \tag{22}$$